POI Detection of High-Rise Buildings Using Remote Sensing Images: A Semantic Segmentation Method Based on Multitask Attention Res-U-Net

Bingnan Li, Jiuchong Gao, Shuiping Chen, Samsung Lim, *Senior Member, IEEE*, and Hai Jiang

***Abstract*— A point-of-interest (POI) represents a specific point location that may be useful or interesting for people, and therefore, each and every building footprint in a topographic map can be recognized as a POI. Automatic extraction of building footprints using remote sensing images has become a challenging and important research topic, which is in demand for urban planning and development. Extensive studies have explored a variety of semantic segmentation methods using deep learning algorithms to achieve better performance in building footprint extraction; however, the existing algorithms were shown to have some limitations, which lead to poor segmentation results. Building roofs were recognized as building footprints in the previous studies. This is prone to error especially for high-rise buildings due to different sensor view angles. In this article, we propose a multitask Res-U-Net model with an attention mechanism for the extraction of the building roofs and the whole building shapes from remote sensing images and then use an offset vector method to detect the footprints of the high-rise buildings based on the boundaries of the corresponding building roofs and shapes. We also apply the online food delivery (OFD) data to parse the POI name of every building footprint. Several strategies are also developed in combination with the proposed model, including data augmentation and postprocessing. We conduct numerical experiments using real data of remote sensing images and OFD historical order data. Results demonstrate that our proposed model achieves a total F1-score of 77.05% and an intersection over union (IoU) of 63.55% in terms of the building roof segmentation, and an overall F1-score of 79.02% and an IoU of 66.05% for the whole building shape segmentation, which both achieve the best performance among all baseline models.**

***Index Terms*— Attention mechanism, building footprint extraction, online food delivery (OFD), remote sensing images, semantic segmentation.**

I. Introduction

AS THE mobile Internet has grown, online food delivery (OFD) services have become increasingly popular in our everyday lives [1]. According to Stock Apps, in 2020, the global OFD market reached $136.4 billion in revenue, an increase of 27% year-over-year, and the number of users reached 1.46 billion, a 25% increase from 2019 [2]. China is leading the way in terms of market size and has achieved great success [3]. Take Meituan, one of the world's largest food delivery platforms as an example. In 2020, total revenues reached 114.8 billion Yuan, an increase of about 18% year-over-year, and the number of users reached around 460 million, a 13% increase from 2018 [4].

OFD platforms rely heavily on accurate point-of-interest (POI) information in their operations. A POI is defined as a specific point location that may be useful or interesting for people, and a region-of-interest (ROI) is the POI's boundary [5], e.g., a residential building footprint. OFD platforms are concerned with two key properties associated with a POI, that is, its name and spatial boundary. Spatial boundaries of POIs are stored as vector formats, which can be easily used in geospatial analysis and improve the service efficiency of the OFD industry.

In Fig. 1, we give an example of POIs in a residential complex. In this figure, the dashed border represents the spatial boundary of an area-of-interest (AOI), which is a gated residential complex. There are 30 buildings in the AOI, and gray polygons represent the footprints of these buildings. For each building footprint, its centroid and building number are also shown in the figure. Each building is recognized as a POI, its spatial boundary is the polygon of the building footprint, and its name is the building number.

An OFD platform uses POI information in many different ways.

1) When a customer places an order, the OFD platform relies on the POI information to resolve the delivery address. Based on the global positioning system (GPS) coordinates of customers and the boundaries of nearby POIs, the platform would suggest possible POI names to assist customers to pinpoint their exact locations, which is critical to ensure timely delivery. For example, a customer is located in the red point, which is shown in Fig. 1; when he/she tries to choose the delivery address, the OFD platform can suggest relevant POI names in a list (e.g., Building #07, Building #08, Building #09, and so on) based on the point's GPS

Manuscript received January 16, 2022; revised April 12, 2022; accepted May 6, 2022. Date of publication May 11, 2022; date of current version May 31, 2022. This work was supported by the Tsinghua University—Meituan Joint Institute for Digital Life. *(Corresponding author: Hai Jiang.)*

Bingnan Li and Samsung Lim are with the School of Civil and Environmental Engineering, University of New South Wales, Sydney, NSW 2052, Australia (e-mail: bingnan.li@unsw.edu.au; s.lim@unsw.edu.au).

Jiuchong Gao and Shuiping Chen are with Meituan, Beijing 100020, China (e-mail: gaojiuchong@meituan.com; chenshuiping@meituan.com).

Hai Jiang is with the Department of Industrial Engineering, Tsinghua University, Beijing 100084, China (e-mail: haijiang@tsinghua.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3174399

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

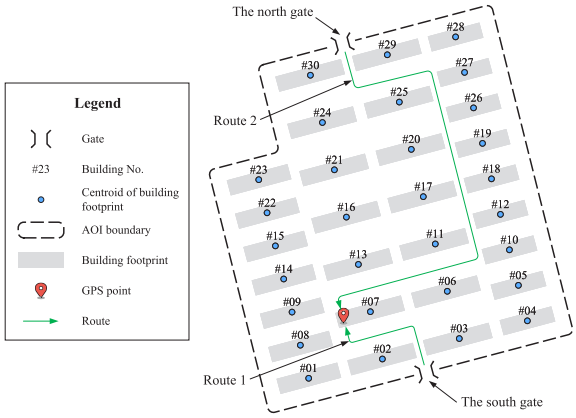


Fig. 1. Example of POIs in a residential complex.

coordinates, which can assist customers to find the right POIs of their delivery addresses.

2) The OFD platform can plan a more efficient route based on the accurate location, and riders, as known as food delivery men, can find the destination efficiently and accurately. For instance, the gated residential complex contains 30 buildings in Fig. 1; if the OFD platform only plans a route to the gate of the residential complex, then riders have to find the destination building by their own familiarity with this place. The optimal route to the customer's place of the red point should be Route 1, while, without accurate POI information, the OFD platform may plan a route to the wrong gate (i.e., the north gate), and riders may find the destination of Building #07 by Route 2, which is very inefficient and time-consuming.

ROI mining techniques are designed to detect the boundaries of POIs. Existing ROI mining techniques are based on three primary approaches: predefined shapes, density-based clustering, and grid-based aggregation. Generally, the boundary of a POI is defined as a cluster, which is given by a geographically shaped convex polygon using the convex hull of the geotagged records (e.g., geotagged Flickr photos) that include a given textual description, such as a POI name within a circle to be optimized [6]. These existing methods highly rely on accurate GPS coordinates, while the accuracy of indoor GPS satellite signals is relatively low, and high-rise buildings can also block or degrade GPS satellite signals of mobile devices, which may cause users to locate wrong locations. Therefore, these methods cannot extract POI boundaries accurately.

As remote sensing technologies mature, the capability to detect the physical boundaries of ground objects has been enhanced significantly. Thus, remote sensing technologies are seen as conventional and even important methods for detecting building footprints [7], which are also regarded as boundaries of POIs. In this study, we extract spatial boundaries of POIs based on remote sensing images. For example, different buildings are located in a residential complex, every one of them can be recognized as a specific POI, and we aim to identify the spatial boundary and the name of each specific POI based on buildings from remote sensing images and OFD data.

Remote sensing images are now getting popular and broadly used in a variety of geoscience applications. Building extraction is advantageous for urbanization planning, disaster management, and environmental management [8], [9]. Because of the diversity of colors, sizes, shapes, and materials of buildings in different areas, and the similarity between buildings and backgrounds or other objects, developing accurate and robust extraction methods of buildings has become a challenging problem. For most existing studies, remote sensing data focus on low-rise buildings, and therefore, building footprints can be generally extracted from the "footprint" of the roof [10]–[12]. However, for high-rise buildings, due to the different view angles of remote sensing sensors, the distance between the polygons of the building roof and the building footprint on remote sensing images can be very large and changes with time. In this case, the polygon of the building roof cannot be regarded as the polygon of the building footprint. Besides building footprints cannot be extracted directly, because most of them are covered by building roofs and building surfaces on remote sensing images, based on data analysis and calculation, we found that, in most cases, the building footprint has the same shape as the building roof, except for a certain offset in position. Therefore, we design multitask deep learning models to extract the boundaries of the building roofs and the whole building shapes, and then, an offset vector method is applied to the boundaries of the building roofs, which can help to get the boundaries of the building footprints.

In this research, we aim to tackle the challenge faced by semantic segmentation of building footprints for high-rise buildings based on remote sensing images. We propose a multitask Res-U-Net model with an attention mechanism to extract the boundaries of building roofs and shapes simultaneously and then use an offset vector method to detect the actual spatial boundaries of the building footprints based on the spatial boundaries of the building roofs and shapes. After detecting the spatial boundaries of the building footprints, parsing of POI names is processed by the OFD dataset. We conduct numerical experiments using remote sensing data from Google Earth and the OFD dataset from the Meituan platform. Experimental results indicate that the proposed method successfully extracts the boundaries of the building roofs and shapes, and improves the total F1-score by 1.78% and 3.31% for the building roof segmentation and the whole building shape segmentation, respectively. The offset vector method helps to obtain the spatial boundaries of building footprints, and the OFD dataset helps to parse POI names.

In this article, we propose a multitask Res-U-Net model with an attention mechanism to extract both the building roofs and shapes of high-rise buildings from remote sensing images. Our proposed model obtains an overall F1-score of 77.05% and intersection over union (IoU) of 63.55% in terms of the building roof segmentation, and an overall F1-score of 79.02% and IoU of 66.05% for the whole building shape segmentation, which both achieve the best performance among baseline models. This article's main contributions can be summed up as follows.

1) To the best of our knowledge, we are the first to investigate the building footprints extraction of high-rise buildings based on remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

2) We propose a novel multitask Res-U-Net model with an attention mechanism for semantic segmentation of the building roofs and shapes. Using the proposed model, the building roofs and the whole building shapes are extracted simultaneously. Compared with the best performing baseline model, the proposed model improves the total F1-score by 1.78% and IoU by 0.49% in terms of the building roof segmentation, and the total F1-score by 3.31% and IoU by 3.03% for the whole building shape segmentation.
3) Most of the existing studies extract building roofs as building footprints, while, in our study, we introduce an offset vector method to extract the building footprints based on boundaries of the building roofs and the whole building shapes for high-rise buildings.
4) Instead of detecting spatial boundaries of the building footprints, our research also parses POI names based on the OFD dataset.

The rest of this article is arranged as follows. In Section II, the related work of POI detection and semantic segmentation of building footprints is described. Section III introduces the architecture of the multitask Res-U-Net model with attention mechanism, the offset vector method, and the name parsing of POIs. In Section IV, we perform numerical experiments to describe the experimental results of our proposed model. We discuss and analyze the extraction results of the building roofs and the whole building shapes based on different models and the ablation study in Section V. Section VI sums up the conclusions of this study and describes the possible future expectations.

II. Related Work

A. POI Detection From VGI Data

Volunteered geographic information (VGI) represents the use of digital devices for the creation, assembly, and dissemination of geographic data voluntarily provided by individuals [13]. VGI data offer an additional data source for a wide range of geoscience applications. Mobile devices with built-in GPS sensors help users share their real-time locations on social media. Existing techniques to find ROIs highly rely on VGI data and can be grouped into three approaches.

For the approach of predefined shapes, ROIs are represented by predefined shapes. In [14], popular touristic routes were extracted using circular ROIs based on geotagged Flickr photos. More precisely, circles are utilized to convert a trajectory of GPS points into an ROI sequence. The approach of density-based clustering uses a set of geolocations to cluster ROIs. DBSCAN [15] is regarded as one of the most commonly used algorithms of density-based clustering for geospatial data. Hu *et al.* [16] applied the DBSCAN clustering algorithm to identify urban ROIs based on Flickr geotagged photos, understand ROIs by extracting unique text tags and favorite images, and analyzed the spatiotemporal dynamics along with perceptions of ROIs. The approach of grid-based aggregation splits the research area into a grid, and ROIs are extracted based on the aggregation of grid cells. In [17], the EXPO 2015 area was divided into a regular grid, and pavilions were represented by the related grid cells of POIs, which was used to discover the behavior and movement patterns of users within the exhibition.

B. Semantic Segmentation of Building Footprints

Over the past few decades, with the growth of remote sensing sensors, remote sensing data paint a detailed picture of the urban environment and make the classification and extraction of buildings and other artificial objects possible [18].

Buildings represent an important part of the city. Automatic and accurate extraction of building boundaries can benefit urbanization planning, disaster management, and environmental management and has been extensively researched over decades [8], [11], [12], [19]. It is seen as the issue of detecting and extracting building areas from images based on techniques of image processing and computer vision [10], [20].

Many studies of building extraction are on the base of traditional image processing methods. Belgiu and Dragut [21] proposed and compared the supervised and unsupervised methods of segmentation in combination with the algorithm of random forest for extracting buildings from remote sensing data. Chen *et al.* [22] proposed the indices of edge patterns and shadow lines as candidates using segmentation models, and different classifiers of machine learning were employed to identify buildings. Ok *et al.* [23] developed a new method of fuzzy landscape generation, which was used to detect the building's directional space relation and shade for automatic building extraction.

Recently, deep learning has been broadly used in different remote sensing applications, i.e., object detection [24], [25], scene classification [26], land use mapping [27], and so on. Deep learning is a specific branch of machine learning, which uses a multiple layer structure to progressively extract higher level features from raw inputs [28]. The deep neural network (DNN) combines operations of linear and nonlinear for encoding deep features of inputs [11].

The convolutional neural network (CNN) is a class of DNN with multiple connected convolutional layers for feature extraction with different levels [29]. CNN is able to achieve better performance, particularly in the field of image processing [30]. As deep learning technology develops, many novel technologies of semantic segmentation are developed. The CNN's structure with multiple layers is able to learn image features from different levels automatically, which is useful to classify images and promote the growth of semantic segmentation. However, image classification of most CNN models is the patch-based method, and pixels are classified by analyzing the area surrounding each pixel. This method is relatively inefficient since patches are overlapping [31].

Chen *et al.* [32] proposed a fully convolutional network (FCN) to address the problem of semantic image segmentation, which could categorize each pixel based on abstract features. In comparison to CNN, FCN is able to deal with images of multiple sizes and improve processing efficiency. Based on the better performance, varieties of methods were developed on the base of FCN. Ronneberger *et al.* [31] proposed an FCN-based U-Net structure, which achieved high

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

performance on medical image segmentation. Chen *et al.* [33] proposed the structure of DeepLab, which utilized operations of deconvolution and upsampling, and expanded deep CNN. Both these two methods focus on the study of improving FCN to achieve a higher performance since the outcome of FCN is tough.

Then, more models with the structure of encoder–decoder are proposed, which enhance the upsampling part [34]. Noh *et al.* [35] proposed the structure of DeconvNet, which converted the process of upsampling to a deconvolution process. CEDN, proposed by Yang *et al.* [36], was developed on the base of encoder–decoder architecture and used for the detection of edges. Badrinarayanan *et al.* [37] proposed the structure of SegNet, which deleted layers with full connection and used max-pooling upsampling. In SegNet, every pooling layer can output two feature maps, namely, the following layer and the decoder of upsampling. Marmanis *et al.* [38] applied FCN and SegNet to detect boundaries by semantic image segmentation, and this model achieved high performance of the segmentation. Shi *et al.* [39] developed a novel change detection network known as DSAMNet, which integrated the convolutional block attention module for more discriminative features both spatialwise and channelwise to achieve fine-grained bitemporal change detection. He *et al.* [40] developed an urban tree-specific subpixel mapping architecture with a deep learning approach to generate a 2-m fine-scale urban tree cover products from 10-m Sentinel-2 images. Deep learning models can achieve better performance than basic machine learning models; therefore, recent studies on semantic image segmentation mostly focus on deep learning networks and following with some optimization operations to achieve good outcomes [41], [42].

According to recent studies, deep learning-based semantic segmentation models are also used to extract buildings based on remote sensing images [11], [12], [43]–[45]. For instance, Maggiori *et al.* [46] used FCN models to achieve the semantic segmentation of buildings from multispectral remote sensing data. Xu *et al.* [47] proposed a building extraction model on the base of the Res-U-Net structure in combination with guided filters. Sahu and Ohri [48] adopted the standard U-Net architecture with batch normalization layers to extract buildings from remote sensing images. The results were then fed into a postprocessing pipeline to separate the connected buildings and converted them into vector formats. Guo *et al.* [43] proposed a multiloss neural network with an attention block based on U-Net, which could extract buildings with high accuracy. The attention block shows clear advantages in extracting features and detecting edges of images. Chen *et al.* [49] developed a shifted windows transformer as a backbone for building extraction from satellite remote sensing images. These studies show the excellent performance of semantic segmentation models of building extraction.

For extensive studies of building footprint extraction, they recognized building roofs as building footprints, which can lead to errors for high-rise buildings due to different sensor view angles of the satellites. Our proposed method can solve this problem. Our method is different from them in various aspects. First, instead of detecting POI boundaries by VGI data, our approach tries to extract building footprints to represent POI boundaries and parse POI names from the OFD dataset. Second, most of the existing studies extract building roofs as building footprints, while, in our study, we introduce an offset vector method to extract building footprints based on the boundaries of the building roofs and the whole building shapes. Third, existing public datasets only label building footprints, while, in this research, we labeled both the building roofs and the whole building shapes based on remote sensing images to meet our research objective. Fourth, an end-to-end deep CNN, that is, a multitask Res-U-Net model with an attention mechanism, is proposed for high-rise building footprint segmentation at pixel level with remote sensing images. Finally, in the proposed model, the attention mechanism shows obvious advantages in extracting features and detecting edges of images, the multitask strategy helps to extract the building roofs and the whole building shapes together, and the encoder is shared by both the decoders.

III. Methodology

This section will detail the proposed multitask Res-U-Net model with an attention mechanism for remote sensing image segmentation. We first describe the regular U-Net model, and two enhanced components, including the ResNet block and the attention mechanism, then describe the overall structure of our proposed model and postprocessing, and, finally, followed by the offset vector method and name parsing of POIs.

A. U-Net

Ronneberger *et al.* [31] proposed the U-Net model, which is essentially a variant of FCN. U-Net is a type of deep CNN model containing connected convolutional layers and deconvolutional layers for the segmentation of biomedical images. U-Net has a symmetrical structure of the left encoder and the right decoder. For the left-hand side, spatial features are extracted based on inputs. For the right-hand side, segmentation maps are built from the extracted spatial features. The decoder part is designed to build the segmentation map based on the extracted spatial features. The decoder is similar in structure of FCN using the combination of multiple convolutional layers. More precisely, the encoder is constitutive of a series of blocks with operations of downsampling, and every block consists of repeated operations of two 3×3 convolutions and an operation of 2×2 max-pooling with stride 2. After each operation of downsampling, the number of filters of the convolutional layers will be doubled. Finally, the encoder part uses operations of two 3×3 convolutions to connect with the decoder part.

On the contrary, the decoder includes a series of blocks for operations of upsampling, which are used to build segmentation images. The decoder uses an operation of 2×2 deconvolution to upsample the feature map, which is produced by the encoder. Zeiler *et al.* [50] proposed the deconvolutional layer, which includes the operation of transposed convolution and halves the number of output filters, followed by a series of blocks for operations of upsampling, which contains two operations of 3×3 convolution and an operation of deconvolution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

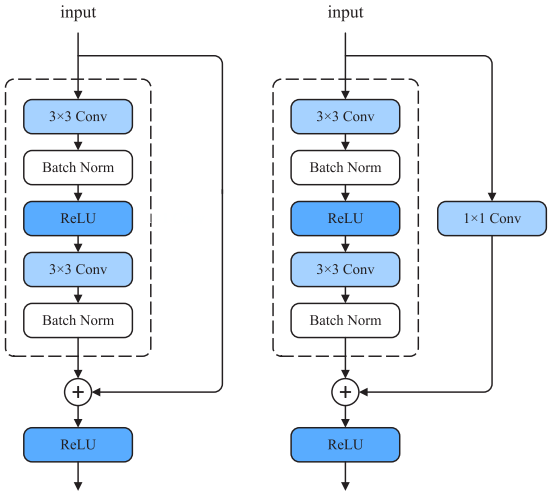


Fig. 2. ResNet block. [(Left) Identity shortcut. (Right) Projection shortcut].

Then, the last layer is a 1×1 convolutional layer, which is used to output the segmentation results. The activation function of the output layer is the Sigmoid function, and all other layers adopt rectified linear unit (ReLU) function. The definitions of these two activation functions are given as follows:

$$\text{Sigmoid function: } f(x) = \frac{1}{1+e^{-x}} \tag{1}$$

$$\text{ReLU function: } f(x) = \max(0, x). \tag{2}$$

B. Residual Network

The residual network (ResNet) [51] is a branch of DNN and is proposed to address the problem of degradation in optimization. To address such a problem, He *et al.* [51] proposed a deep residual learning framework where sets of layers fit a residual map in contrast to other architectures where each layer fits the entire underlying map.

Let $H(x)$ represent the last desired mapping; then,

$$F(x) = H(x) - x \tag{3}$$

where x denotes the input image and $F(x)$ represents the residual mapping (the portion within the dotted-line box in Fig. 2). In a feedforward DNN, (3) is called as a shortcut connection. More specifically, (3) can be rewritten as

$$y_i = F(x_i, \{W_i\}) + x_i \tag{4}$$

where y_i and x_i stand for, respectively, the output and input vectors of the ith set of stacked layers, and $F(x_i, \{W_i\})$ is the shortcut connection to be learned. Because the shortcuts should have an equal size with x, we can map these as identity or as a linear projection through short connections to match the dimensions. Fig. 2 illustrates an instance of ResNet block, which is utilized to construct the network. In the ResNet block, every convolutional layer is connected to a batch normalization [52]. In our proposed model, the number of channels of x_i should be changed in the ResNet block; therefore, the ResNet block with projection shortcut is applied to our model.

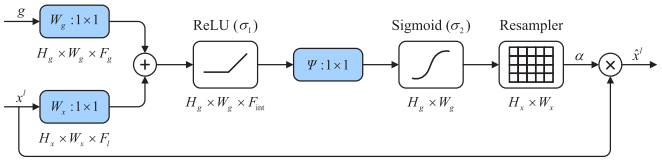


Fig. 3. Schematic of AG. Input features (x^l) are scaled based on attention coefficient (α) calculated from AG. Spatial areas are chosen by analyzing activations together with contextual information from the gating signal (g). Grid resampling of attention coefficient is used to make it have the same height and width with x^l.

C. Attention Gate

The attention gate (AG) for image analysis was introduced by Oktay *et al.* [53]. AG can retrieve information with high dimensions and exclude information from backgrounds through the attention mechanism, which is illustrated in Fig. 3. The convolutional layer provides a representation of images with high dimensions (x^l) by a layer of local information. In the end, pixels are separated based on semantic differences of target features in space with high dimensions. The feature map is acquired through a linear conversion and the ReLU function at the output of the layer l, and the ReLU function can be represented as $\sigma_1(x^l_{i,c}) = \max(0, x^l_{i,c})$, where i and c denote spatial and channel dimensions, respectively, and then followed by an operation of 1×1 convolution with just one filter and the activation function of Sigmoid. Based on this process, all values of the feature map are scaled to the interval [0, 1].

Attention coefficients, $\alpha_i \in [0, 1]$, identify prominent image areas and mitigate the feature response to keep activations related to the particular task. The result of AGs are calculated as $\hat{x}^l_{i,c} = x^l_{i,c} \cdot \alpha^l_i$. A unique scale attention value is calculated for every vector of pixels $x^l_i \in \mathbb{R}^{F_l}$, where F_l represents the number of corresponding feature maps of layer l. As shown in Fig. 3, a gating vector $g_i \in \mathbb{R}^{F_g}$ is utilized for every pixel i to identify focus areas. The gating vector includes contextual information to mitigate responses of lower level features [54].

D. Model Architecture

The architecture of multitask Res-U-Net model with attention mechanism is an encoder–decoder network on the base of the U-Net structure, with a shared encoder path to the left and two structurally similar decoder paths to the right, namely: 1) *shared encoder*; 2) *building roof decoder*; and 3) *building shape decoder*. The outline of the proposed model is illustrated in Fig. 4.

1) Shared Encoder: The encoder is utilized to encode features on different levels and then shared with two decoders. More precisely, the encoder path contains repeated operations of downsampling, which doubles the number of feature channels and halves the size of feature maps in every step. Every operation of downsampling is preceded by a ResNet block, which is mentioned in Section III-B.

2) Building Roof Decoder: This is to predict the segmentation of the building roofs from the information of the shared encoder (i.e., the building roof decoder task). More precisely, every decoder block is symmetrical to the encoder, and in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

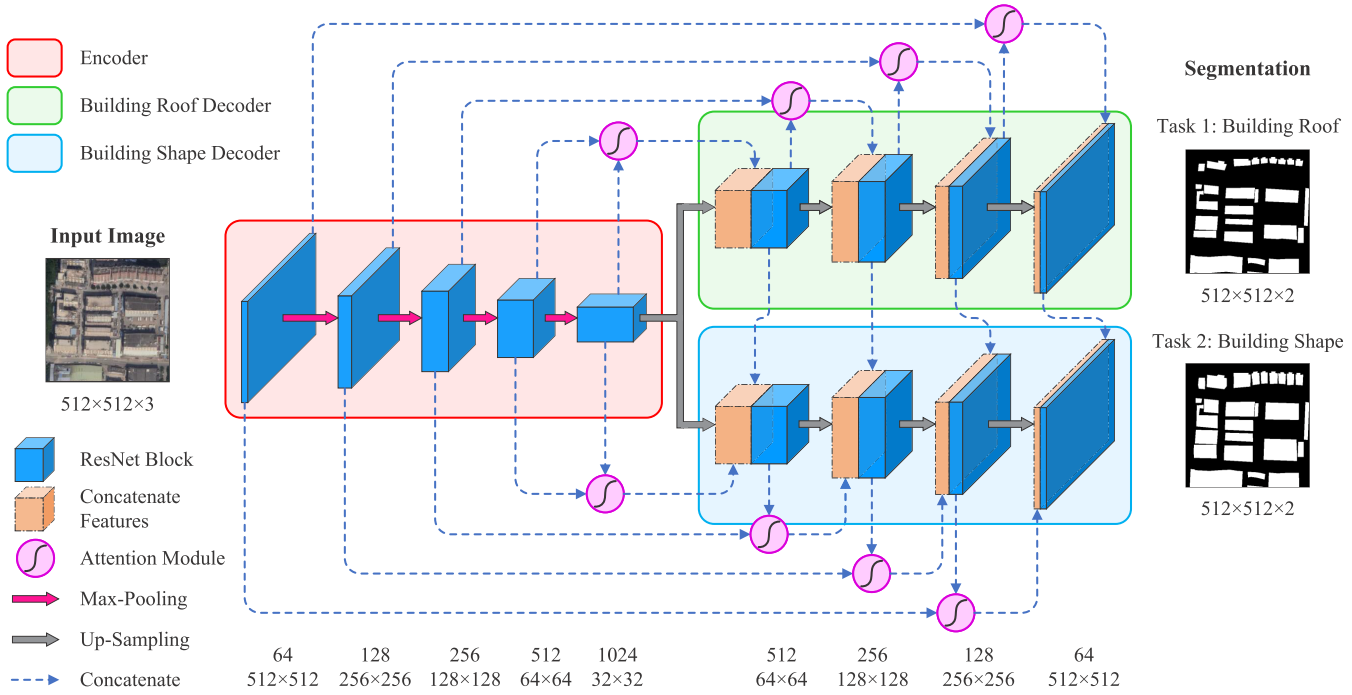


Fig. 4. Architecture of the proposed multitask Res-U-Net model with an attention mechanism. Every blue cuboid represents a ResNet block. The number of channels and x–y size are denoted at the bottom of the figure. The attention module filters the features propagated via the skip connections, and dotted orange cuboids represent concatenate features. The arrows denote the different operations.

every decoder layer, features of the corresponding encoder layer are concatenated with an attention module, which enables the maintenance of multiscale features, the enhancement of important channel features, and the weakening of unimportant channel features.

3) Building Shape Decoder: This is to predict the segmentation of the whole building shapes based on the information of the shared encoder and the building roof decoder. This is similar to the building roof decoder except that, when merging the information in every layer, we concatenate not only the information of the shared encoder but also the information of the building roof decoder. The building roof is always a part of the whole building shape; hence, the learned building roof information would help with the segmentation prediction task of the whole building shape.

E. BCE and Dice Loss

Loss functions define how models calculate the overall error of predicted results and ground truth. The choice of loss functions can directly affect the learning process and final results of models. Even a properly constructed model can be affected by an improper loss function. In this research, extraction of the building roof and the whole building shape can be defined as two problems of binary segmentation in which the loss function of binary cross-entropy (BCE) is often used, as indicated by (5). However, remote sensing images of buildings exist an unbalanced issue between pixels of buildings and backgrounds in which the loss function of BCE tends to be trapped at the local minimum, and the network is prone to predict a good loss value of the background and does not learn the feature representation of the minor class. One solution is adding weights to these classes during the loss calculation, which introduces extra hyperparameters that require careful adjustment. The other way is to select a more balanced function, e.g., BCE–dice loss.

The definition of BCE–dice loss is the summation of BCE loss and dice loss, which combines the benefits of both functions. The BCE loss can express the erroneous classification well and is easy to calculate the gradient in spite of the defect mentioned above. The dice loss function, as indicated by (6), is constructed on a dice coefficient, which measures the overlapping area of the predicted result and the ground truth. If they match each other more, the dice coefficient is closer to 1, driving the value of dice loss closer to 0. On the contrary, if every pixel is predicted with the wrong value, the largest value of dice loss is 1. However, BCE loss performs better at this point, as the loss function of BCE can generate values far greater than 1, which speeds up the progress of optimization. The BCE–dice loss is indicated by (7).

The prediction and ground truth of the building roof are denoted as $\hat{\mathbf{Y}}_r \in \mathbb{Z}_2^{I\times J}$ and $\mathbf{Y}_r \in \mathbb{Z}_2^{I\times J}$, respectively, where $\mathbb{Z}_2$ means the set of binary values $\{0, 1\}$. The prediction and ground truth of the whole building shape are denoted as $\hat{\mathbf{Y}}_s \in \mathbb{Z}_2^{I\times J}$ and $\mathbf{Y}_s \in \mathbb{Z}_2^{I\times J}$, respectively. Specifically, the BCE loss of the prediction $\hat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$ is

$$\mathcal{L}_{\mathrm{BCE}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = -\sum_{i}^{I}\sum_{j}^{J}\left[Y_{ij}\log\left(\hat{Y}_{ij}\right) + \left(1 - Y_{ij}\right)\log\left(1 - \hat{Y}_{ij}\right)\right] \quad (5)$$

where I and J are pixel numbers in height and width directions, respectively, i and j are the pixel indices, $Y_{ij} \in \{0, 1\}$ is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

the pixel label class, and $\hat{Y}_{ij} \in [0, 1]$ is the probability of the pixel being predicted as the positive category. In this research, pixels of the building roof and the whole building shape are the positive category, and pixels of the background are negative category.

The dice loss of the prediction $\hat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$ is

$$\mathcal{L}_{\text{Dice}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = 1 - \frac{2\sum_i^I \sum_j^J \hat{Y}_{ij} Y_{ij} + \epsilon}{\sum_i^I \sum_j^J \hat{Y}_{ij} + \sum_i^I \sum_j^J Y_{ij} + \epsilon} \tag{6}$$

where ϵ is utilized to guarantee the stability of the loss function by avoiding the numeric problem of being divided by 0.

The BCE–dice loss of the prediction $\hat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$ is

$$\mathcal{L}_{B-D}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = (1-\lambda)\mathcal{L}_{\text{BCE}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) + \lambda \mathcal{L}_{\text{Dice}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) \tag{7}$$

where λ denotes the hyperparameter that balances BCE loss and dice loss. If the boundary information of the building roof and the whole building shape is more complex, then the value of λ needs to be decreased; in contrast, if the boundary information of the building roof and the whole building shape are more regular, then the value of λ needs to be increased [43]. In our research area, the boundary information of the building roof and the whole building shape is more regular, and the value of λ is set to 0.8.

We minimize the hybrid loss of the building roof prediction and the whole building shape prediction by gradient descent

$$\mathcal{L}(\boldsymbol{\theta}) = (1-\varphi)\mathcal{L}_{B-D}\left(\hat{\mathbf{Y}}_r, \mathbf{Y}_r\right) + \varphi \mathcal{L}_{B-D}\left(\hat{\mathbf{Y}}_s, \mathbf{Y}_s\right) \tag{8}$$

where $\boldsymbol{\theta}$ represents a collection of all trainable parameters and φ denotes the hyperparameter that balances the hybrid loss of the building roof prediction and the whole building shape prediction. If the boundary information of the building roof is more important than that of the whole building shape, then the value of φ needs to be decreased; in contrast, if the boundary information of the whole building shape is more important than that of the building shape, then the value of φ needs to be increased [55]. In this research, the boundary information of both the building shape and the whole building shape is very important, and the value of φ is set to 0.5.

F. Postprocessing

The output obtained from the proposed multitask Res-U-Net model with attention mechanism is in binary format, in which every pixel indicates the category of the building roof or the background for task 1 and indicates the category of the whole building shape or the background for task 2.

Since raster data for segmentation of the building roofs and the whole building shapes are not very useful for spatial analysis and spatial calculation, an operation of postprocessing was applied in this study. In postprocessing, raster data are taken as input, and spatial boundaries of the building roofs and the whole building shapes are generated as vector format after noise removal and separation of connected objects.

1) Noise Removal: For input images, pixels of the building roof and the whole building shape are represented in white, and pixels of the background are represented in black, which are shown in the segmentation part of Fig. 4. In order to remove the noises of detected building roofs and building shapes, we apply common morphological operations to the binary images. We use the operation of erosion first and then the operation of dilation, which can remove small noises automatically.

2) Distance Transform: In order to separate the connected building roofs or building shapes, we assume that the area of the connected part is smaller than the area of the building roof or the whole building shape itself. Based on this fact, distance transformation [48] is applied to the binary images. Therefore, the connected part is given a smaller weight in comparison with the building roof or the whole building shape itself.

3) Local Maxima: After the operation of distance transformation, the local maxima method [48] is introduced. Since the connected part is smaller than the building roof or the whole building shape itself, finding local maxima can make sure that it is within the building roof or the whole building shape and not within the connected part. The local maximum point is used as the input sink of the watershed segmentation.

4) Watershed Segmentation: Watershed segmentation is widely used in image segmentation. This method treats the image like a topographic map, with the brightness of each point representing its height, and finds the lines that run along the tops of ridges [56]. Local maxima are used as the sink point, and the negative of the distance transform is regarded as the cost map. This method can be used to separate the connected binary blobs with high efficiency.

5) Vectorization and Smoothing: Raster images obtained from watershed segmentation are vectorized based on libraries of OpenCV2[1] and GDAL/OGR.[2] In the raster images, every pixel contains the geographical information, and raster images are converted into vector files with correct overlaying (Shapely[3] library). When converting data from raster into the vector, results have redundant vertices and noises. The algorithm of Douglas and Peucker [57] is used to simplify vector files. This algorithm makes it possible to maintain the general shape of the geometry when redundant vertices are simplified. Moreover, basic properties (e.g., area and perimeter of spatial boundaries) are automatically recorded.

6) Polygon Matching: After getting polygons of the building roofs and the whole building shapes, the process of polygon matching can help us detect each pair of the building roof and the whole building shape. In order to solve this problem, the Rtree data structure is introduced. The main idea of the Rtree data structure is to group nearby objects and represent them with their minimum bounding rectangle in the next higher level of the tree. This data structure can effectively accelerate the nearest neighbor search for spatial data. In this research, for every polygon of the building roof, it will recall all polygons of the building shapes that intersect it based on the Rtree data structure. Among all candidate polygons of the

[1] https://opencv.org/opencv-2-4-8/

[2] https://gdal.org/

[3] https://shapely.readthedocs.io/en/stable/manual.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

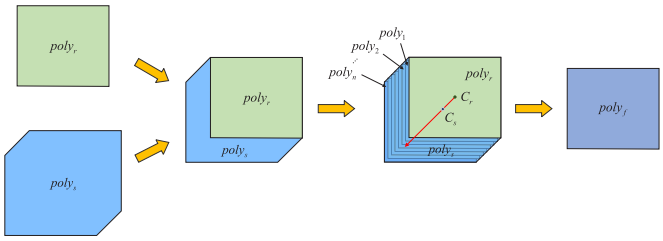


Fig. 5. Working principle diagram of the offset vector method.

building shapes, only the one that has the largest overlap area with the polygon of the building roof is recognized as the corresponding polygon of the building shape.

G. Offset Vector Method

Based on the postprocessing in Section III-F, raster images can be converted to vector polygons. For every remote sensing image, two sets of vector polygons can be generated, i.e., a set of vector polygons of the building roofs $\Psi_r = \{\text{poly}_1^r, \text{poly}_2^r, \ldots, \text{poly}_n^r\}$, and a set of vector polygons of the whole building shapes $\Psi_s = \{\text{poly}_1^s, \text{poly}_2^s, \ldots, \text{poly}_n^s\}$. Then, based on the offset vector method, a set of vector polygons of the building footprints $\Psi_f = \{\text{poly}_1^f, \text{poly}_2^f, \ldots, \text{poly}_n^f\}$ can be calculated.

As noted in Section I, in most instances, the building footprint has the same shape as the building roof, except for a certain offset in position. After getting the vector polygons of the building roof and the whole building shape, the offset vector helps us to move the polygon of the building roof to the position of the building footprint. Suppose that poly_i^r is the ith polygon of Ψ_r, poly_i^s is the corresponding ith polygon of Ψ_s, poly_i^f is set to the corresponding ith polygon of Ψ_f, and $\overrightarrow{v_i}$ is the offset vector from poly_i^r to poly_i^f. Then, C_i^r represents the centroid of poly_i^r, and C_i^s represents the centroid of poly_i^s. Both C_i^r and C_i^s lie on the offset vector $\overrightarrow{v_i}$; therefore, $\overrightarrow{v_i}$ can be calculated by C_i^r and C_i^s. When the polygon of poly_i^r moves along the vector of $\overrightarrow{v_i}$, the critical position is defined as poly_i^r, which is about to move out poly_i^s, and the position of poly_i^r is exactly the polygon of poly_i^f.

To understand this method easier, a flowchart is shown in Fig. 5. The green polygon is denoted by poly_r, which represents the polygon of the building roof. The blue polygon is denoted by poly_s, which represents the polygon of the whole building shape. Then, poly_r and poly_s are overlaid together with the same coordinate system. The third step shows how the offset vector method works. C_r denotes the centroid of poly_r, and C_s denotes the centroid of poly_s. Based on C_r and C_s, the offset vector is created, which is shown as the red vector in the figure. Then, poly_r is moved along the offset vector with the same incremental step size every time, and a set of polygons is generated, i.e., $\Phi_r = \{\text{poly}_1, \text{poly}_2, \ldots, \text{poly}_n\}$. Then, the polygon is about to move out poly_s, which is defined as the polygon of the building footprint, which is denoted as poly_f. Finally, poly_f is calculated by the offset vector method based on poly_r and poly_s.

Algorithm 1 Offset Vector Method

Input: $poly_r, poly_s, step, threshold$
Output: $poly_f$

```
1  begin
2      x_r ← poly_r.centroid.x
3      y_r ← poly_r.centroid.y
4      x_s ← poly_s.centroid.x
5      y_s ← poly_s.centroid.y
6      vector_x ← x_s − x_r
7      vector_y ← y_s − y_r
8      area_inter ← area(poly_r ∩ poly_s)
9      i ← 0
10     poly_i ← poly_r
11     area_inter_i ← area(poly_i ∩ poly_s)
12     while area_inter_i/area_inter ≥ threshold do
13         i ← i + 1
14         offset_x ← vector_x × step × i
15         offset_y ← vector_y × step × i
16         poly_i ← translate(poly_r, offset_x, offset_y)
17         area_inter_i ← area(poly_i ∩ poly_s)
18     end
19     poly_f ← poly_i
20     return poly_f
21 end
```

The offset vector method is also implemented by Algorithm 1. For the input of this algorithm, poly_r denotes the polygon of the building roof, and poly_s is the polygon of the corresponding building shape. The parameter of step represents the incremental step size, and threshold refers to a value to determine whether the polygon is about to move out poly_s or not. These two parameters can be set to different values for different situations. For the offset vector method, on Lines 2–5, we retrieve the centroid of poly_r, i.e., (x_r, y_r), and the centroid of poly_s, i.e., (x_s, y_s). On Lines 6 and 7, the offset vector is calculated based on the two centroids, which is denoted as $(\text{vector}_x, \text{vector}_y)$. When poly_r is moved along the offset vector, a series of polygons will be generated, and poly_i denotes the ith polygon of them. For poly_i, area_inter_i refers to the area of intersection between poly_i and poly_s, and area_inter means the area of intersection between poly_r and poly_s. Then, on Lines 12–18, a while loop is used to find the polygon of the building footprint. If the value of $\text{area_inter}_i/\text{area_inter}$ is greater than or equal to threshold, the while loop continues; otherwise, the while loop stops. In each iteration, poly_i moves the distance of one step size (step) along the offset vector on Lines 13–16. The while loop stops when the value of $\text{area_inter}_i/\text{area_inter}$ is less than threshold. Finally, the last ploy_i is set to poly_f on Line 19, and poly_f is returned as the polygon of the building footprint. The detailed proof procedure of the offset vector method is presented in the Appendix.

Based on the offset vector method, polygons of the building footprints can be calculated from the corresponding polygons of the building roofs and the whole building shapes. For extensive studies of building footprint extraction [10]–[12], [58], the building roofs are recognized as the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

building footprints. However, in this research, the offset vector method is used to detect building footprints based on building roofs and the whole building shapes. The limitation of this method is that it cannot be applied to those buildings where the shapes of the building roofs and the building footprints are very different, such as buildings with pointed roofs and special form buildings.

H. Name Parsing of POIs

Based on the process of Section III-G, the spatial boundaries of POIs are detected from the building roofs and the whole building shapes, and the next step is to parse names of POIs. The historical order dataset from the Meituan platform consists of rich geospatial information about customers and riders. Suppose that poly_i^f is one polygon of the building footprint from Ψ_f in Section III-G, and $O_i = \{o_1, o_2, \ldots, o_m\}$ is a set of orders with customers' geographic coordinates located in the boundary of poly_i^f. Based on the natural language processing technology, names of POIs can be parsed from delivery addresses of O_i. Then, $N_i = \{n_1, n_2, \ldots, n_m\}$ denotes the set of parsed names of POIs from O_i. GPS drift refers to the difference between the real location and the location recorded by a GPS receiver, and it makes some GPS points locate in the wrong locations. Therefore, multiple POI names (i.e., N_i) can be found in poly_i^f, but only the most frequent name in N_i is recognized as the name of poly_i^f. Based on this method, POI names can be easily parsed from the historical order dataset.

IV. Numerical Experiments

In this section, we first outline the dataset of the study. Then, it is followed by the data augmentation and experimental settings. Evaluation metrics and experimental results of the proposed model are described in the end.

A. Data Descriptions

For existing publicly available remote sensing building datasets [59]–[61], they focus on low-rise buildings, and the building outline can be generally recognized as the building footprint. However, in our study, we focus on high-rise buildings, and the building roofs and the whole building shapes should be extracted separately. Therefore, we did a lot of work on labeling the data for training and testing. The data were obtained from remote sensing images of urban areas of Shenzhen, China, in Google Earth. These remote sensing images include three bands, and the format is GeoTIFF with geographic information. In this study, we collected 108 remote sensing images sized 1024 × 1.024 pixels. For every image, we manually outlined all the building roofs and the whole building shapes in it, respectively, which is illustrated in Fig. 6. In this study, remote sensing images were labeled by the VGG image annotator (VIA) tool [62]. For both the building roofs and the whole building shapes, annotated images were first saved as the JavaScript Object Notation (JSON) format and then were converted into binary images with only black and white. Finally, every remote sensing image was split into four images with the size of 512 × 512, and the same operation was also performed on the binary images. The dataset used in this research is available at https://github.com/BuildingFootprint/POI-Detection-of-High-Rise-Buildings-Using-Remote-Sensing-Images-Dataset.

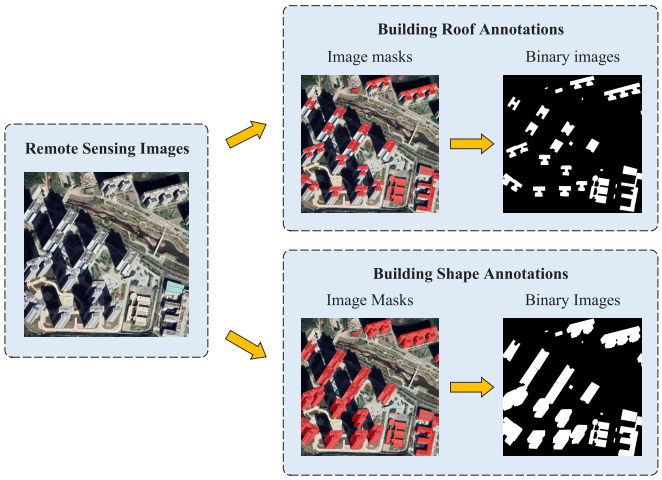


Fig. 6. Annotation diagram of remote sensing images.

TABLE I
Dataset Division

	Numbers of Samples	Numbers of buildings
Training dataset	256	8,553
Validation dataset	88	2,891
Test dataset	88	3,067

In the experiment, the whole dataset covers 14 511 building objects; polygons of the building roofs and the whole building shapes are seen as the ground truth for training and evaluating models. The dataset is split into three sets: the training dataset, the validation dataset, and the test dataset with ratios of 60%, 20%, and 20%. The dataset division is illustrated in Table I.

B. Data Augmentation

Deep CNNs need a large number of training data, which may be unavailable at the training stage. Data augmentation of images is critical to make the network invariant and robust, and to avoid overfitting if the training samples are not large enough. If data augmentation is applied, the model always sees the different sets of images during every iteration. In this study, various methods of data augmentation are applied as follows. For every pair of input images and ground-truth binary images, they were flipped horizontally or vertically randomly based on a probability of 0.5. Images were offset to left, right, up, or down with a random number of pixels. Moreover, images were rotated with a random angle and a random center, and zoomed in or out based on a random scale factor. Finishing the transformation, the rest pixel values of images were set to 0 s. Then, these augmented images were fed into models as original training images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

C. Experimental Settings

The multitask Res-U-Net model with attention mechanism was built by Keras based on the Tensorflow backend. In this study, the size of input images for every experiment was $512 \times 512 \times 3$. Our proposed model and other baseline models were all trained on Nvidia Tesla M60 (four GPUs with 16-GB memory each).

All hyperparameters used in these experiments were optimal parameters in comparison with repeated test results. During the training stage, the Adam algorithm was chosen as the optimizer, and the initial learning rate was set with 10^{-4}. Weights and parameters of models were initialized by the Glorot normal initializer [63]. For all models, the epoch was set to 50, and the minibatch size was set to 4.

D. Evaluation Metrics

In order to quantify the performance of models, precision, recall, F1-score, and IoU are utilized as qualitative metrics. In semantic segmentation, precision represents the proportion of correctly classified positive pixels in all pixels predicted as positive. The recall is defined as the proportion of correctly classified positive pixels in all true positive pixels. F1-score is the harmonic mean of precision and recall. IoU is used to evaluate the accuracy based on the segment level. The definition of the four metrics is represented as follows:

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{9}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{10}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times N_{\text{TP}}}{2 \times N_{\text{TP}} + N_{\text{FP}} + N_{\text{FN}}} \tag{11}$$

$$\text{IoU} = \frac{N_{\text{TP}}}{N_{\text{FP}} + N_{\text{TP}} + N_{\text{FN}}} \tag{12}$$

where N_{TP} is the number of overlapping building roof (or building shape) pixels in predicted and ground-truth images (true positive). N_{FP} is the number of building roof (or building shape) pixels in the predicted image but not in the ground-truth image (false positive). N_{FN} is the number of building roof (or building shape) pixels in the ground-truth image but not in the predicted image (false negative).

E. Experimental Results

In this study, we divided experiments into three stages. The first stage was to extract polygons of the building roofs and the whole building shapes based on the multitask Res-U-Net model with an attention mechanism. The second stage was to postprocess polygons of the building roofs and the whole building shapes and detect polygons of the building footprints based on the offset vector method. The third stage was to combine POI information with the building footprints based on historical OFD orders.

Detailed results for the proposed model are given in Table II. For every evaluation metric, the performance of the whole building shape segmentation is better than that of the building roof segmentation. F1-score and IoU are two important indices for evaluating the performance of semantic segmentation. For the building roof segmentation, F1-score and IoU are 77.05% and 63.55%, respectively. For the whole building shape segmentation, F1-score and IoU are 79.02% and 66.05%, respectively. Overall, F1-score and IoU are 78.04% and 64.80%, respectively.

Fig. 7 illustrates a sample diagram of the segmentation results based on the test dataset. We take three cases (i.e., Case 1, Case 2, and Case 3) as examples to illustrate the results. For every case, we use two rows to show the results of the building roof segmentation and the whole building shape segmentation, respectively. In order to show the results clearly,

TABLE II
EVALUATION RESULTS OF THE PROPOSED MODEL

Index	Building Roof	Building Shape	Overall
Precision	75.45%	77.35%	76.40%
Recall	79.92%	82.05%	80.99%
F1-score	77.05%	79.02%	78.04%
IoU	63.55%	66.05%	64.80%

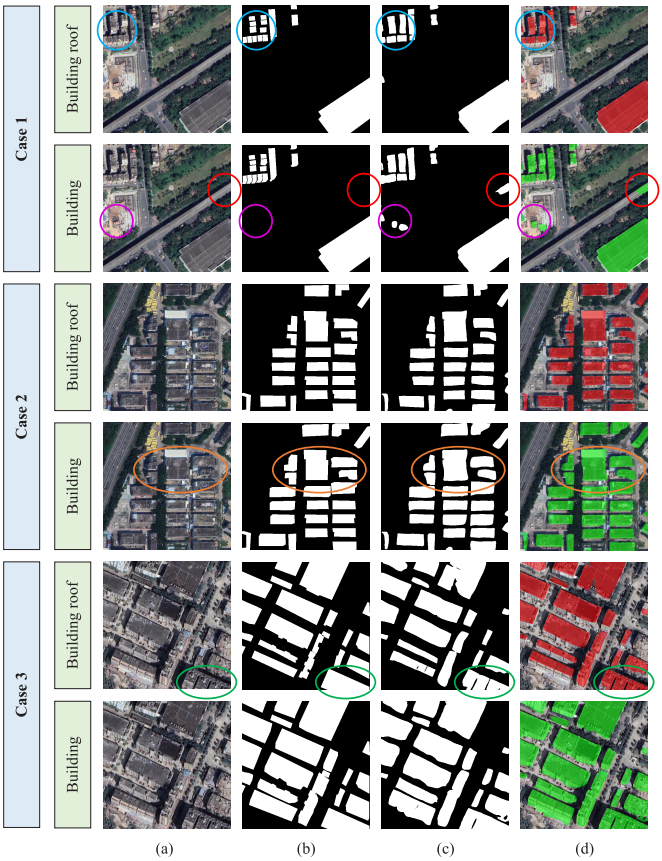


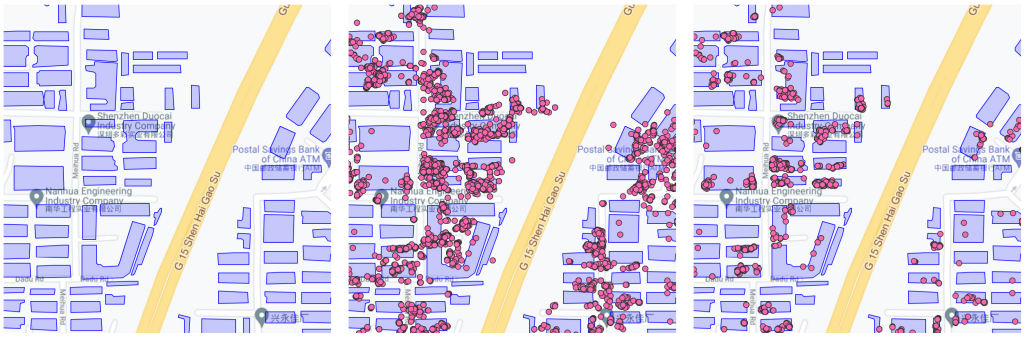
Fig. 7. Results of the building roof extraction and the whole building shape extraction on the test dataset. (a) Original remote sensing images. (b) Ground-truth images of the building roof and the whole building shape. (c) Predicted binary images of the building roof and the whole building shape based on the proposed model. (d) Overlapping display of the predicted masks with the original remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 8. Spatial boundaries of the building roofs and footprints. (a) Spatial boundaries of the building roofs. (b) Spatial boundaries of the building footprints. (c) Spatial boundaries of the building roofs and footprints.



(a) (b) (c)

Fig. 9. Name parsing of POIs. (a) Spatial boundaries of the building footprints. (b) Spatial boundaries of the building footprints and GPS points located in the research area. (c) Spatial boundaries of the building footprints and GPS points located in them.

we use red masks to represent the predicted building roofs and green masks to represent the predicted whole building shapes.

Comparing the predicted segmentation results with the ground truth, we can still find some problems. In Case 1, the narrow gaps between closely neighbored buildings are annotated in the ground truth, but our proposed model is incapable of expressing such fine details and separating these kinds of buildings, e.g., segmentation results in the blue circles. Within the red circles, a part of the road is wrongly recognized as a building, and within the magenta circles, buildings under construction are detected, but they are not included in the ground truth. In Case 2, compared with the ground truth, polygons of predicted results always have curve boundaries, which are shown in the orange ellipses. In the green ellipses of Case 3, we can see that shadows of buildings are excluded in the segmentation results, but they are included in the ground truth.

After detecting polygons of the building roofs and the whole building shapes, we need to match every building roof polygon to the corresponding building shape polygon. In order to increase the matching efficiency, we introduced an R-tree data structure, which is utilized for the effective storage of spatial data indices. Then, based on the offset vector method mentioned in Section III-G, polygons of the building footprints can be calculated from polygons of the building roofs and the whole building shapes.

An example is illustrated in Fig. 8 to explain this process. In Fig. 8(a), the red line represents the spatial boundary of the building roof, and the background is the remote sensing image. The green line in Fig. 8(b) represents the spatial boundary of the building footprint. In Fig. 8(c), the spatial boundaries of the building roofs and footprints are overlaid together, and most of the building roofs have an offset distance from the corresponding building footprints. However, some spatial boundaries of building roofs and building footprints overlap with each other, which means that the offset distance of these cases is 0 m. After calculation, in this research area, the largest offset distance is 9.08 m, and the average offset distance is 3.70 m. The value of the offset distance varies with the different view angles of remote sensing sensors and different heights of buildings.

Then, the name parsing of POIs is processed based on OFD data, and the whole procedure is shown in Fig. 9. Fig. 9(a) illustrates building footprints in the research area, and every polygon represents a building footprint. In Fig. 9(b), customers' GPS locations of historical OFD order data are overlaid on the map, and every pink point represents the customer's GPS location of order. All GPS locations located in the research area are shown in this figure. Then, in the next step, only orders with GPS locations located in polygons of the building footprints are kept, which is shown in Fig. 9(c), and other orders are removed. Finally, the most frequent POI name of orders located in every building footprint is recognized as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE COMPARISON OF SEMANTIC SEGMENTATION BASED ON FCN-8S, U-NET, RES-U-NET, SEGNET, DEEPLABV3+, DECONVNET, AND THE PROPOSED MODEL IN TERMS OF PRECISION (%), RECALL (%), F1-SCORE (%), AND IOU (%), WHERE THE BOLD VALUES REPRESENT THE BEST FOR CORRESPONDING METRICS

Model	Building Roof				Building Shape			
	Precision	Recall	F1-score	IoU	Precision	Recall	F1-score	IoU
FCN-8s	70.68%	77.31%	72.92%	58.69%	70.28%	77.85%	73.19%	58.72%
U-Net	72.50%	75.84%	73.22%	59.71%	74.60%	79.35%	75.54%	62.15%
Res-U-Net	76.96%	73.71%	74.13%	60.74%	75.48%	79.76%	76.34%	63.48%
SegNet	76.53%	76.21%	75.70%	63.24%	77.61%	76.51%	76.49%	64.11%
DeepLabV3+	68.40%	69.57%	68.04%	52.78%	75.21%	62.75%	67.41%	51.48%
DeConvNet	72.95%	62.29%	65.89%	50.44%	76.24%	67.26%	72.02%	56.19%
Our Model	75.45%	79.92%	**77.05**%	**63.55**%	77.35%	82.05%	**79.02**%	**66.05**%

TABLE IV

QUANTITATIVE COMPARISON OF ABLATION STUDY IN TERMS OF PRECISION (%), RECALL (%), F1-SCORE (%), AND IOU (%), WHERE THE BOLD VALUES REPRESENT THE BEST FOR CORRESPONDING METRICS

Model	Building Roof				Building Shape			
	Precision	Recall	F1-score	IoU	Precision	Recall	F1-score	IoU
Model-A	75.35%	77.29%	74.33%	61.05%	77.94%	79.43%	76.76%	64.16%
Model-R	75.34%	75.24%	74.26%	60.02%	76.51%	77.67%	76.05%	62.47%
Model-M	75.35%	76.94%	74.87%	61.04%	75.41%	81.03%	76.95%	64.30%
Our Model	75.45%	79.92%	**77.05**%	**63.55**%	77.35%	82.05%	**79.02**%	**66.05**%

the name of the building footprint. As shown in the figure, an issue is that some building footprints cannot recall any orders, and an effective method to solve this problem is to collect OFD order data for a long period of time.

V. DISCUSSION

A. Comparison With Different Methods

In order for the comparison with the performance of the proposed model, we use six mainstream remote sensing image segmentation models, including FCN-8s [64], U-Net [31], Res-U-Net [47], SegNet [37], DeepLabV3+ [34], and DeConvNet [35], as these networks are very representative and effective for semantic segmentation based on remote sensing images.

As mentioned in Section IV-D, precision, recall, F1-score, and IoU are metrics that can illustrate the performance of semantic image segmentation. Quantitative results of four metrics are provided in Table III. The bold values of F1-score and IoU represent the best results among these models. Overall, the proposed model outperforms other deep CNNs in both the building roof segmentation and the whole building shape segmentation. For the proposed model, the F1-score and IoU of the building roof segmentation are 77.05% and 63.55%, respectively, which are the highest among all models; these two values of the whole building shape segmentation are 79.02% and 66.05%, respectively, which are also the highest among all models. Moreover, the classification accuracy of the whole building shape is better than that of the building roof. Res-U-Net achieves the best precision (76.96%) in the building roof segmentation, which is higher than that of our proposed model (75.45%). Compared with our proposed model, SegNet achieves a relatively high performance of precision in both the building roof segmentation and the whole building shape segmentation, which are 76.53% and 77.61%, respectively. For both the building roof segmentation and the whole building shape segmentation, the performance of the F1-score and IoU of SegNet is the highest among all the baseline models, with F1-score and IoU of 75.70% and 63.24% for the building roof segmentation and with F1-score and IoU of 76.49% and 64.11% for the whole building shape segmentation. Compared with other baseline models, DeepLabV3+ and DeConvNet have a relatively worse performance of F1-score and IoU for both the building roof segmentation and the whole building shape segmentation.

To sum up, the proposed model has been compared with different baseline models and proved to be more effective. As illustrated in Table III, compared with baseline models, the performance of the proposed model is the best.

B. Ablation Study

Apart from the comparison results of different models, we also focus on the efficiency of every component of our proposed model. Therefore, the ablation study of our proposed model is conducted to explore how these components impact the performance of semantic segmentation. To make sure an equitable experimental comparison, the ablation study is conducted in the exact same environment as our major experiments mentioned in Section IV-C. We perform the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

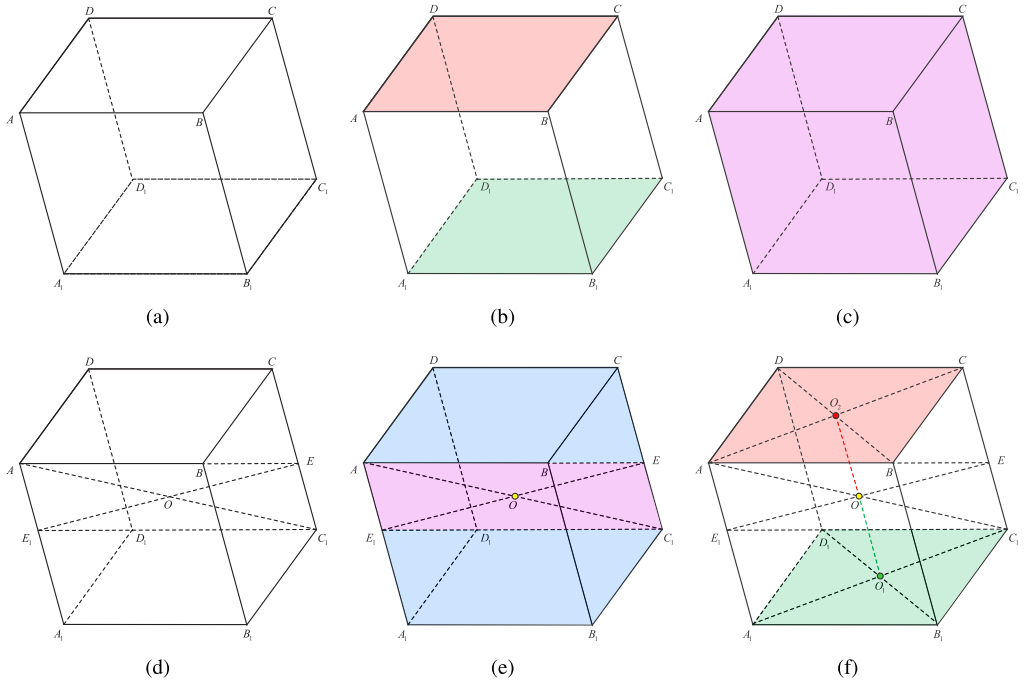


Fig. 10. (a) Parallelepiped of the projected building. (b) Polygons of the building roof and the building footprint. (c) Polygon of the whole building shape. (d) Auxiliary lines are added for calculation. (e) Calculation of the centroid of the whole building shape. (f) Proof of centroids of the building roof, the building footprint and the whole building shape all lie on the same vector.

ablation study on three variations of our proposed model: model without the attention module, model without the ResNet block, and model without the multitask strategy. The results of the ablation study are summarized in Table IV, where Model-A represents the model without the attention module, i.e., the model of multitask Res-U-Net; Model-R represents the model without the ResNet block, i.e., model of multitask U-Net with attention mechanism; and Model-M represents the model without the multitask strategy, i.e., model of Res-U-Net with an attention mechanism. From the table, we can observe that the proposed model achieves the best performance for both the building roof segmentation and the whole building shape segmentation compared with Model-A, Model-R, and Model-M. Therefore, the attention module, the ResNet block, and the multitask strategy all play significant roles in our proposed model and help improve the performance of the semantic segmentation compared with U-Net.

VI. Conclusion and Future Research Directions

In this research, we proposed a multitask Res-U-Net model with an attention mechanism to increase the performance of semantic segmentation of the building roofs and the whole building shapes, and applied a sequence of postprocessing methods and the offset vector method to detect building footprints of high-rise buildings from remote sensing images. Based on the historical order dataset from the OFD platform known as the Meituan platform, we also identified the POI names of the detected building footprints. After designing the attention module, the ResNet block, and the multitask strategy in the structure of the network, the capability to enhance the model's performance and optimize the boundaries of segmentation results can be achieved. Compared with the existing models, our model achieved the best performance in terms of both F1-score and IoU. The results showed that the attention mechanism, the ResNet block, and the multitask strategy could effectively increase the amount of information and improve the predictive capability of the model.

Even though the proposed model works excellent as a supervised model, it relies on a mass of manually labeled data. Further studies are required to facilitate manual annotation. Potential orientations for exploration include semisupervised semantic segmentation for adversarial learning, which can use unlabelled data to generate self-learning content signals to divide the network. To further scientific knowledge, we will continue to work toward this direction and explore the hybrid model to produce significant semantic explanations.

Appendix
Proof of the Offset Vector Method

This appendix presents the detailed proof procedure of the offset vector method. For the offset vector method, the hypothesis of this research is that the polygon of the building roof has the same shape as the polygon of the building footprint in the real world. Generally, the distance between the orbit of a satellite and the Earth's surface is thousands of kilometers, while the height of a building is only tens of meters. Therefore, the shape of the building roof and the corresponding building footprint in the remote sensing image can be recognized as the same.

To prove that the offset vector is correct, we suppose that the building is a parallelepiped. When the building is projected onto a plane, its whole shape is shown in Fig. 10(a). This figure should be seen as a 2-D image; thus, $AA_1 // BB_1 // CC_1 // DD_1$, $AB // dc // D_1C_1 // A_1B_1$. In Fig. 10(b), the light red polygon (the polygon $ABCD$) represents the polygon of the building roof, and the light green polygon

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

(the polygon $A_1B_1C_1D_1$) represents the polygon of the building footprint. In Fig. 10(c), the light purple polygon (the polygon $A_1B_1C_1CDA$) represents the polygon of the whole building shape. We need to prove that the centroid of the polygon $ABCD$, the centroid of the polygon $A_1B_1C_1D_1$, and the centroid of the polygon $A_1B_1C_1CDA$ all lie on the same vector. Both the polygon $ABCD$ and the polygon $A_1B_1C_1D_1$ are parallelograms; therefore, the centroids are the intersection of the diagonals. Then, we need to calculate the centroid of the polygon $A_1B_1C_1CDA$.

To calculate the centroid of the polygon $A_1B_1C_1CDA$, we need to add some auxiliary lines. In Fig. 10(d), extend the line segment AB to the line segment CC_1, and the intersection point is set to E. Then, extend the line segment C_1D_1 to the line segment AA_1, and the intersection point is set to E_1. In Fig. 10(e), the polygon $A_1B_1C_1CDA$ is divided into three polygons with different colors, i.e., the polygon $AECD$, the polygon $A_1B_1C_1E_1$, and the polygon E_1C_1EA. It is easy to prove that the polygon E_1C_1EA is a parallelogram; thus, the centroid of it is the intersection of its diagonals, which is the point O. It is also easy to prove that the polygon $AECD$ is equivalent to the polygon $A_1B_1C_1E_1$, and they are centrosymmetric with respect to the point O. Therefore, the centroid of the polygon $AECD$ and the polygon $A_1B_1C_1E_1$ is the point O, and we can prove that the point O is the centroid of the polygon $A_1B_1C_1CDA$. In Fig. 10(f), the point O_2 is the centroid of the polygon $ABCD$, the point O_1 is the centroid of the polygon $A_1B_1C_1D_1$, and then, we need to prove that the point O_2, the point O, and the point O_1 lie on the same line. Because the point O_2 is the midpoint of the line segment AC, the point O is the midpoint of the line segment E_1E, the point O_1 is the midpoint of the line segment A_1C_1, and $AA_1 // CC_1$; therefore, $OO_2 // CC_1$, $OO_1 // CC_1$, and the point O, the point O_1, and the point O_2 all lie on the line segment O_1O_2. Finally, we prove that the centroid of the polygon $ABCD$, the centroid of the polygon $A_1B_1C_1D_1$, and the centroid of the polygon $A_1B_1C_1CDA$ all lie on the same vector.

For most of the residential buildings, their shapes are close to the cuboid. Therefore, the offset vector method can be applied to these buildings.

REFERENCES

[1] J. Zheng, L. Wang, S. Wang, Y. Liang, and J. Pan, "Solving two-stage stochastic route-planning problem in milliseconds via end-to-end deep learning," *Complex Intell. Syst.*, vol. 7, no. 3, pp. 1207–1222, Feb. 2021.

[2] J. Kranjec. (2021). *Online Food Delivery Market to hit $151.5b in Revenue and 1.6b Users in 2021, a 10% Jump in a Year*. [Online]. Available: https://stockapps.com/blog/2021/01/05/online-food-delivery-market-to-hi%t-151-5b-in-revenue-and-1-6bn-users-in-2021-a-10-jump-in-a-year

[3] C. Li, M. Mirosa, and P. Bremer, "Review of online food delivery platforms and their impacts on sustainability," *Sustainability*, vol. 12, no. 14, p. 5528, Jul. 2020.

[4] Meituan. (2021). *Meituan Announces Financial Results for the Year Ended December 31, 2020*. [Online]. Available: https://www.prnewswire.com/news-releases/meituan-announces-financial-re%sults-for-the-year-ended-december-31-2020-301256643.html

[5] L. Belcastro, F. Marozzo, D. Talia, and P. Trunfio, "G-RoI: Automatic region-of-interest detection driven by geotagged social media data," *ACM Trans. Knowl. Discovery From Data*, vol. 12, no. 3, pp. 1–22, Jun. 2018.

[6] C. Tran, D. D. Vu, and W.-Y. Shin, "An improved approach for estimating social POI boundaries with textual attributes on social media," *Knowl.-Based Syst.*, vol. 213, Feb. 2021, Art. no. 106710.

[7] X. Zhang, Y. Sun, A. Zheng, and Y. Wang, "A new approach to refining land use types: Predicting point-of-interest categories using Weibo check-in data," *ISPRS Int. J. Geo-Inf.*, vol. 9, no. 2, p. 124, Feb. 2020.

[8] Y. Sun, X. Zhang, X. Zhao, and Q. Xin, "Extracting building boundaries from high resolution optical images and LiDAR data by integrating the convolutional neural network and the active contour model," *Remote Sens.*, vol. 10, no. 9, p. 1459, Sep. 2018.

[9] S. Shrestha and L. Vanneschi, "Improved fully convolutional network with conditional random fields for building extraction," *Remote Sens.*, vol. 10, no. 7, p. 1135, 2018.

[10] W. Li, C. He, J. Fang, J. Zheng, H. Fu, and L. Yu, "Semantic segmentation-based building footprint extraction using very high-resolution satellite images and multi-source GIS data," *Remote Sens.*, vol. 11, no. 4, p. 403, 2019.

[11] K. Zhao, M. Kamran, and G. Sohn, "Boundary regularized building footprint extraction from satellite images using deep neural network," 2020, *arXiv:2006.13176*.

[12] A. Ziaee, R. Dehbozorgi, and M. Döller, "A novel adaptive deep network for building footprint segmentation," 2021, *arXiv:2103.00286*.

[13] M. F. Goodchild, "Citizens as sensors: The world of volunteered geography," *GeoJournal*, vol. 69, no. 4, pp. 211–221, Aug. 2007.

[14] E. Spyrou and P. Mylonas, "Analyzing Flickr metadata to extract location-based information and semantically organize its photo content," *Neurocomputing*, vol. 172, pp. 114–133, Jan. 2016.

[15] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. KDD*, vol. 96, 1996, pp. 226–231.

[16] Y. Hu, S. Gao, K. Janowicz, B. Yu, W. Li, and S. Prasad, "Extracting and understanding urban areas of interest using geotagged photos," *Comput., Environ. Urban Syst.*, vol. 54, pp. 240–254, Nov. 2015.

[17] E. Cesario *et al.*, "Analyzing social media data to discover mobility patterns at EXPO 2015: Methodology and results," in *Proc. Int. Conf. High Perform. Comput. Simulation (HPCS)*, Jul. 2016, pp. 230–237.

[18] J. Hui, M. Du, X. Ye, Q. Qin, and J. Sui, "Effective building extraction from high-resolution remote sensing images with multitask driven deep neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 786–790, May 2018.

[19] L. Li, J. Liang, M. Weng, and H. Zhu, "A multiple-feature reuse network to extract buildings from remote sensing imagery," *Remote Sens.*, vol. 10, no. 9, p. 1350, Sep. 2018.

[20] E. Maggiori *et al.*, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Oct. 2016.

[21] M. Belgiu and L. Dragut, "Comparing supervised and unsupervised multiresolution segmentation approaches for extracting buildings from very high resolution imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 96, pp. 67–75, Oct. 2014.

[22] R. Chen, X. Li, and J. Li, "Object-based features for house detection from RGB high-resolution images," *Remote Sens.*, vol. 10, no. 3, p. 451, Mar. 2018.

[23] A. O. Ok, C. Senaras, and B. Yuksel, "Automated detection of arbitrarily shaped buildings in complex environments from monocular VHR optical satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1701–1717, Mar. 2013.

[24] P. Ding, Y. Zhang, W.-J. Deng, P. Jia, and A. Kuijper, "A light and faster regional convolutional neural network for object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 141, pp. 208–218, Jul. 2018.

[25] W. Li, R. Dong, H. Fu, and L. Yu, "Large-scale oil palm tree detection from high-resolution satellite images using two-stage convolutional neural networks," *Remote Sens.*, vol. 11, no. 1, p. 11, 2019.

[26] Y. Liu, Y. Zhong, F. Fei, Q. Zhu, and Q. Qin, "Scene classification based on a deep random-scale stretched convolutional neural network," *Remote Sens.*, vol. 10, no. 3, p. 444, Mar. 2018.

[27] B. Huang, B. Zhao, and Y. Song, "Urban land-use mapping using a deep convolutional neural network with high spatial resolution multispectral remote sensing imagery," *Remote Sens. Environ.*, vol. 214, pp. 73–86, Sep. 2018.

[28] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, Nov. 2015.

[29] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.

[30] M. Alam, J.-F. Wang, C. Guangpei, L. Yunrong, and Y. Chen, "Convolutional neural network for the semantic segmentation of remote sensing images," *Mobile Netw. Appl.*, vol. 26, no. 1, pp. 200–215, Feb. 2021.

[31] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Cham, Switzerland: Springer, 2015, pp. 234–241.

[32] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[33] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[34] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[35] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528.

[36] J. Yang, B. Price, S. Cohen, H. Lee, and M.-H. Yang, "Object contour detection with a fully convolutional encoder-decoder network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 193–202.

[37] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2015.

[38] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2016.

[39] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[40] D. He, Q. Shi, X. Liu, Y. Zhong, and L. Zhang, "Generating 2m fine-scale urban tree cover product over 34 metropolises in China based on deep context-aware sub-pixel mapping network," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102667.

[41] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[42] J. F. Mas and J. J. Flores, "The application of artificial neural networks to the analysis of remotely sensed data," *Int. J. Remote Sens.*, vol. 29, no. 3, pp. 617–663, 2008.

[43] M. Guo, H. Liu, Y. Xu, and Y. Huang, "Building extraction based on U-Net with an attention block and multiple losses," *Remote Sens.*, vol. 12, no. 9, p. 1400, Apr. 2020.

[44] P. Liu *et al.*, "Building footprint extraction from high-resolution images via spatial residual inception convolutional neural network," *Remote Sens.*, vol. 11, no. 7, p. 830, 2019.

[45] Q. Zhu, C. Liao, H. Hu, X. Mei, and H. Li, "MAP-Net: Multiple attending path neural network for building footprint extraction from remote sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6169–6181, Jul. 2021.

[46] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Fully convolutional neural networks for remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5071–5074.

[47] Y. Xu, L. Wu, Z. Xie, and Z. Chen, "Building extraction in very high resolution remote sensing imagery using deep learning and guided filters," *Remote Sens.*, vol. 10, no. 1, p. 144, 2018.

[48] M. Sahu and A. Ohri, "Vector map generation from aerial imagery using deep learning," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. IV-2/W5, pp. 157–162, May 2019.

[49] X. Chen, C. Qiu, W. Guo, A. Yu, X. Tong, and M. Schmitt, "Multiscale feature learning by transformer for building extraction from satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[50] M. D. Zeiler, D. Krishnan, G. W. Taylor, and R. Fergus, "Deconvolutional networks," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2528–2535.

[51] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[52] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[53] O. Oktay *et al.*, "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.

[54] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3156–3164.

[55] S. Ruan *et al.*, "Learning to generate maps from trajectories," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 1, 2020, pp. 890–897.

[56] R. Bandara, "Image segmentation using unsupervised watershed algorithm with an over-segmentation reduction technique," 2018, *arXiv:1810.03908*.

[57] D. H. Douglas and T. K. Peucker, "Algorithms for the reduction of the number of points required to represent a digitized line or its caricature," *Int. J. Geographic Inf. Geovis.*, vol. 10, no. 2, pp. 112–122, 1973.

[58] B. Bischke, P. Helber, J. Folz, D. Borth, and A. Dengel, "Multi-task learning for segmentation of building footprints with deep neural networks," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1480–1484.

[59] V. Mnih, *Machine Learning for Aerial Image Labeling*. Toronto, ON, Canada: Univ. Toronto, 2013.

[60] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.

[61] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[62] A. Dutta and A. Zisserman, "The VIA annotation software for images, audio and video," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 2276–2279.

[63] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.

[64] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.



Bingnan Li received the B.Sc. and M.Sc. degrees in cartography and geographic information systems from East China Normal University, Shanghai, China, in 2011 and 2014, respectively. He is currently pursuing the Ph.D. degree with the School of Civil and Environmental Engineering, University of New South Wales, Sydney, NSW, Australia.

His research interests include deep learning for geospatial data analysis and spatio-temporal data mining.



Jiuchong Gao received the M.Eng. degree in pattern recognition from the Dalian University of Technology, Dalian, China, in 2014.

He is currently an Algorithm Engineer with Meituan, Beijing, China. His research interests include machine learning, natural language processing, and ubiquitous computing.



Shuiping Chen received the B.Eng. degree in industrial engineering and the M.Eng. degree in management science and engineering from Tsinghua University, Beijing, China, in 2011 and 2014, respectively.

He is currently an Algorithm Engineer with Meituan, Beijing. His research interests include spatial–temporal data mining, computer vision, and operations research.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.



Samsung Lim (Senior Member, IEEE) received the B.A. and M.A. degrees in mathematics from Seoul National University, Seoul, South Korea, in 1988 and 1990, respectively, and the Ph.D. degree in aerospace engineering and engineering mechanics from The University of Texas at Austin, Austin, TX, USA, in 1995.

He currently teaches geographic information systems (GIS) at the School of Civil and Environmental Engineering, University of New South Wales (UNSW), Sydney, NSW, Australia. His publications in GIS include spatial and temporal fire occurrence patterns using landscape attributes, such as climate, terrain, and land use to develop a better understanding of wildfire occurrence across Australia, identifying the most significant attributes linked to fire occurrence, effective flood emergency plans and response measures, integration of spatial analysis and an agent-based model for shelter assignment and routing in flood scenarios, and risk analysis methods to flag epidemics for urgent intervention. His publications in light detection and range (LiDAR) and remote sensing include deep learning from multispectral/hyperspectral images, data fusion of LiDAR point clouds and multispectral/hyperspectral images, spatial information extraction from LiDAR data, UAV imagery and aerial/satellite imagery for data segmentation and classification, digital elevation modeling, feature extraction, building edge detection, change detection, and monitoring environmental changes. His main research interests are threefold: GIS, LiDAR, and remote sensing.

Dr. Lim is also a National Delegate of Commission 3, International Federation of Surveyors (FIG), a National Representative of the International Cartographic Association (ICA) Commission on Sensor-driven Mapping, and a Professional Member of the International Association of Geodesy (IAG).



Hai Jiang received the B.Eng. degree in civil engineering from Tsinghua University, Beijing, China, in 2011, and the Ph.D. degree in transportation and logistics from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 2006.

He is currently an Associate Professor with the Department of Industrial Engineering, Tsinghua University. His research is supported by leading companies, such as Didi, China, Ford, USA, AutoNavi, China, and Meituan, China. His teaching and research interests include the modeling and optimization of transportation/ logistics systems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:28:21 UTC from IEEE Xplore. Restrictions apply.